

AI 500 - Foundations of AI

Written Assignment 1

Deadline: 9th November 2024, Saturday

Name: _____

Roll Number: _____

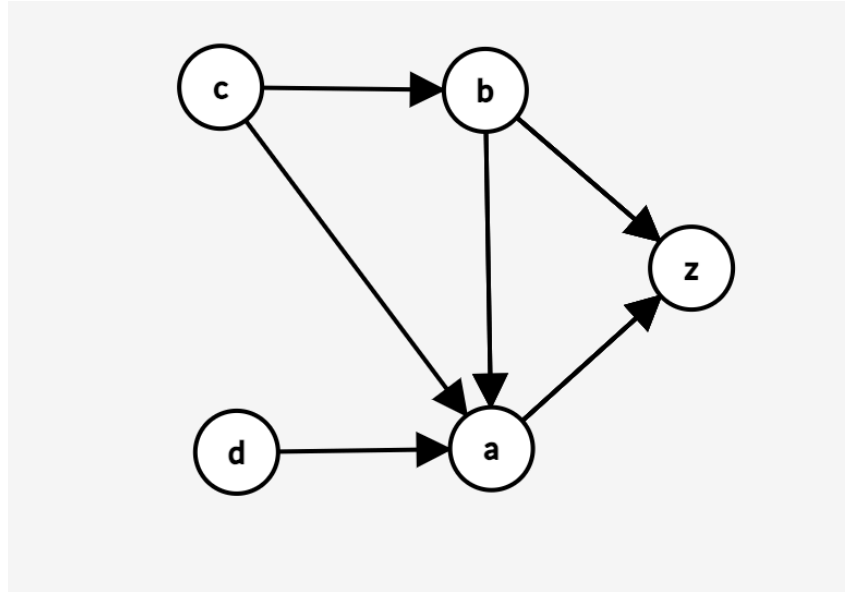
Instructions

- This is an individual assignment. You are expected to solve the questions on your own and not seek help from other students. Please reach out to the TAs if you have any questions regarding the assignment. Plagiarism will not be tolerated.
- You are supposed to print the assignment and attempt the questions in the provided space. It is preferred that you present your solutions in the given boxes. An extra page is given at the end for more work if needed. Do not attach more pages.
- All questions are compulsory.
- A hard copy of the assignment should be submitted in class before the start of the lecture on 9th November 2024, Saturday.
- The late submission policy is the same as for programming assignments, as stated on LMS. 5% deduction per day for up to 5 days.
- This assignment is worth a total of 31 marks. Problem 1 is worth 10 marks, Problem 2 is worth 21 marks.

Best of Luck!

Problem 01 Computational Graphs - 10 marks

Consider the following computational graph



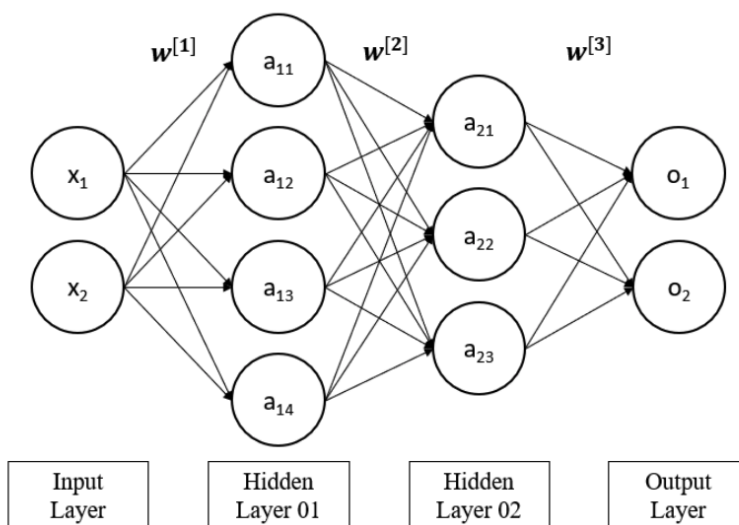
- $a = 3d^3 + 2c + b^2$
- $b = 2c^2$
- $z = 3a + 2b^2$
- where $c = 2$ and $d = 5$

1. Perform a forward pass and compute the value of z , b and a (**3 marks**)

2. Using Chain rule compute the value of dz/dc . Remember direct substitution of values will lead to 0 marks. **(7 marks)**

Problem 02 Back Propagation – 21 marks

Consider the neural network diagram given below:



- $w_{ij}^{[k]}$ represents the value of the weight belonging to the k^{th} layer (shown in the diagram above), between the i^{th} node in the input layer and j^{th} node in the output layer. For example, the weight between the nodes a_{12} and a_{23} will be represented as $w_{23}^{[2]}$.

$w_{11}^{[1]}$	$w_{12}^{[1]}$	$w_{13}^{[1]}$	$w_{14}^{[1]}$	$w_{21}^{[1]}$	$w_{22}^{[1]}$	$w_{23}^{[1]}$	$w_{24}^{[1]}$
0.9	0.75	-0.15	0.33	-0.79	-0.03	0.1	0.02
$w_{11}^{[2]}$	$w_{12}^{[2]}$	$w_{13}^{[2]}$	$w_{21}^{[2]}$	$w_{22}^{[2]}$	$w_{23}^{[2]}$	$w_{31}^{[2]}$	$w_{32}^{[2]}$
-0.58	1.0	-0.8	0.23	0.45	0.05	0.25	-0.65
$w_{33}^{[2]}$	$w_{41}^{[2]}$	$w_{42}^{[2]}$	$w_{43}^{[2]}$	$w_{11}^{[3]}$	$w_{12}^{[3]}$	$w_{21}^{[3]}$	$w_{22}^{[3]}$
-0.09	-0.57	0.51	-0.33	0.97	0.3	-0.01	-0.2
$w_{31}^{[3]}$	$w_{32}^{[3]}$	-----	-----	-----	-----	-----	-----
0.85	0.76	-----	-----	-----	-----	-----	-----

- The activation function for the hidden layers is ***tanh*** and for the output layer it is ***sigmoid***.
- The cost function for the neural network is ***binary cross entropy***.
- The learning rate used for weight updates is ***0.01***.
- The input values for x_1 and x_2 are ***0.5*** and ***0.3*** respectively.
- The actual values (ground truth) for o_1 and o_2 are ***0.4*** and ***0.45*** respectively.
- Assume that the biases are ***0***.

Part A – Forward Pass (4+3+2 marks)

1. Using the appropriate weights and activation function, compute the values of the nodes for the *first hidden layer*.

2. Using the appropriate weights and activation function, compute the values of the nodes for the **second hidden layer**.

- Using the appropriate weights and activation function, compute the values of the nodes for the *output layer*.

Part B – Backward Propagation and Weight Update (3+3+3+3 marks)

- By using partial derivatives, compute the derivative of the error w.r.t. the weights $w_{31}^{[3]}$.
(Hint: You may find it helpful to use the derivatives you computed in Problem 1)

2. By using partial derivatives, compute the derivative of the error w.r.t. the weights $w_{23}^{[2]}$.
(Hint: You may find it helpful to use the derivatives you computed in the previous part)

3. By using partial derivatives, compute the derivative of the error w.r.t. the weights $w_{12}^{[1]}$.
(Hint: You may find it helpful to use the derivatives you computed in the previous part)

4. Use the computed derivatives to update the weights $w_{31}^{[3]}$, $w_{23}^{[2]}$ and $w_{12}^{[1]}$.

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